



MISO DPP 2020 West Area Study Phase 2 Final Report

November 17, 2022

**MISO
720 City Center Drive
Carmel
Indiana 46032
<http://www.misoenergy.org>**

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1. Executive Summary

This report presents the results of a System Impact Study (SIS) performed to evaluate the interconnection of the generators in the DPP 2020 West Area Phase 2 (West Area DPP 2). The study was performed under the direction of MISO and reviewed by an ad hoc study group. The ad hoc study group was formed to review the study scope, methodology, models and results. The ad hoc study group consisted of representatives from the interconnection customers and the following utility companies – American Transmission Company, Ameren, Basin Electric Power Cooperative, CIPCO, Cedar Falls, Dairyland Power Cooperative, Great River Energy, ITC Midwest, Montana-Dakota Utilities, MidAmerican, Minnesota Power, Ottertail Power, SMMPA, and Xcel Energy.

1.1. Project List

The original interconnection requests for DPP 2020 West Area had a total of 30 projects generation projects with a combined nameplate rating of 3492 MW (ERIS) & 2788 MW (NRIS). The detailed list of 2020 West Area DPP Phase 2 is shown below in **Error! Reference source not found.**, and the 2020 West cycle originally kicked off on January 6, 2021.

Project ID	Fuel Type	TO	Service Request Type	Service Request (MW)	POI	Interconnection Self Fund Election
J1494	Battery Storage	Northern States Power (Xcel Energy)	NRIS	50	Chisago 115kV	Yes
J1495	Battery Storage	Northern States Power (Xcel Energy)	NRIS	50	North Rochester 161kV	Yes
J1498	Battery Storage	Northern States Power (Xcel Energy)	NRIS	50	Chanarambie 115 kV Substation	Yes
J1503	Solar	ITC Midwest	NRIS	150	Huntley 345 kV	Yes
J1504	Solar	ITC Midwest	NRIS	150	Huntley 345 kV	Yes
J1520	Solar	ITC Midwest	NRIS	135	Ledyard - Colby 345 kV	Yes
J1528	Battery Storage	Northern States Power (Xcel Energy)	ERIS	0*	King – Eau Claire 345 KV	Yes
J1529	Solar	MidAmerican Energy Company	NRIS	50	Carroll - Auburn 69 kV	Yes
J1530	Solar	MidAmerican Energy Company	NRIS	50	Hastings - Macedonia 69 kV	Yes
J1532	Solar	MidAmerican Energy Company	NRIS	50	Clarinda-Coburg 69 kV	Yes

Project ID	Fuel Type	TO	Service Request Type	Service Request (MW)	POI	Interconnection Self Fund Election
J1534	Solar	Southern Minnesota Municipal Power Agency	NRIS	100	Byron 345kV	No
J1566	Solar	Southern Minnesota Municipal Power Agency	NRIS	150	Rutland Substation 161kV	No
J1572	Hybrid	Northern States Power (Xcel Energy)	NRIS	150	Huntley - Blue Earth 161kV	Yes
J1575	Wind	Otter Tail Power Company	NRIS	70	Warsaw - Donaldson 115 kV	Yes
J1581	Solar	Northern States Power (Xcel Energy)	NRIS	200	Nobles County 345 kV Substation	Yes
J1582	Solar	Northern States Power (Xcel Energy)	NRIS	160	Pine Lake - Three Lakes 115 kV	Yes
J1588	Solar	Northern States Power (Xcel Energy)	NRIS	200	Bison 345kV	Yes
J1590	Wind	MidAmerican Energy Company	NRIS	200	O'Brien - Raun 345 kV	Yes
J1605	Solar	Northern States Power (Xcel Energy)	NRIS	200	Sherburne 345kV	Yes
J1611	Solar	Minnesota Power (Allete Inc)	NRIS	120	Langola 115kV Tap	Yes
J1620	Solar	Northern States Power (Xcel Energy)	ERIS	125	Pipestone to Split Rock 115kV	Yes
J1622	Solar	Otter Tail Power Company	NRIS	100	Harvey-Underwood 230kV	Yes
J1657	Solar	Northern States Power (Xcel Energy)	ERIS	100	Pine Lake - Apple River 161 kV	Yes
J1730	Solar	Northern States Power (Xcel Energy)	NRIS	300	Eau Claire - AS King 345 kV	Yes
J1734	Solar	ITC Midwest	NRIS	150	Wever 161 kV	Yes
J1754	Wind	Great River Energy	NRIS	200	McHenry-Coal Creek Tap 230 kV	No
J1769	Wind	Dairyland Power Cooperative	ERIS	80	J898 POI 161 kV	Yes

Project ID	Fuel Type	TO	Service Request Type	Service Request (MW)	POI	Interconnection Self Fund Election
J1804	Battery Storage	Northern States Power (Xcel Energy)	ERIS	0*	Pine Lake – Three Lakes 115 kV	Yes
J1808	Solar	Dairyland Power Cooperative	ERIS	148	Barron 161 kV	Yes
J1826	Hydro	Northern States Power (Xcel Energy)	ERIS	4	Cannon Falls 69 kV	Yes

*Denotes a Hybrid Application with another DPP 2020 Project



1.2. Project Summary Network Upgrades

J1494			
Network Upgrade	NU Cost Estimate (\$)	J1494 Cost Estimate (\$)	NU Type
J1181 POI-Hazelton 345 Sag Remediation	\$500,000	\$10,122	NRIS
Double Circuit Raun to Sioux City 345 kV	\$35,200,000	\$565,509	SPP
Double Circuit Split Rock to Sioux City 345 kV	\$126,000,000	\$2,120,174	SPP
Double Circuit Raun to S3451 345 kV	\$115,554,682	\$1,974,013	SPP
Rebuild S3451 to S3454 345 kV	\$38,481,000	\$581,925	SPP
Rebuild S3451 to S3459 345 kV	\$23,418,800	\$353,073	SPP
Rebuild S3456 to S3459 345 kV	\$15,444,600	\$232,969	SPP
Total Cost Per Project:		\$5,837,785	
J1495			
Network Upgrade	NU Cost Estimate (\$)	J1495 Cost Estimate (\$)	NU Type
J1181 POI-Hazelton 345 Sag Remediation	\$500,000	\$15,402	NRIS
Double Circuit Raun to Sioux City 345 kV	\$35,200,000	\$416,112	SPP
Double Circuit Split Rock to Sioux City 345 kV	\$126,000,000	\$1,676,327	SPP
Double Circuit Raun to S3451 345 kV	\$115,554,682	\$1,795,401	SPP
Rebuild S3451 to S3454 345 kV	\$38,481,000	\$539,962	SPP
Rebuild S3451 to S3459 345 kV	\$23,418,800	\$315,735	SPP
Rebuild S3456 to S3459 345 kV	\$15,444,600	\$207,397	SPP
Total Cost Per Project:		\$4,966,336	
J1498			
Network Upgrade	NU Cost Estimate (\$)	J1498 Cost Estimate (\$)	NU Type
3rd Nobles 345/115 kV Transformer	\$10,000,000	\$10,000,000	NRIS
J1181 POI-Hazelton 345 Sag Remediation	\$500,000	\$7,437	NRIS
Double Circuit Raun to Sioux City 345 kV	\$35,200,000	\$1,131,672	SPP
Double Circuit Split Rock to Sioux City 345 kV	\$126,000,000	\$4,709,916	SPP



Double Circuit Raun to S3451 345 kV	\$115,554,682	\$3,029,212	SPP
Rebuild S3451 to S3454 345 kV	\$38,481,000	\$886,319	SPP
Rebuild S3451 to S3459 345 kV	\$23,418,800	\$542,811	SPP
Rebuild S3456 to S3459 345 kV	\$15,444,600	\$357,588	SPP
White Swan 250 MVar Reactive Support	\$6,222,750	\$241,044	SPP
Columbus 200 MVar Reactive Support	\$3,788,600	\$104,372	SPP
Total Cost Per Project:		\$21,010,371	
J1503			
Network Upgrade	NU Cost Estimate (\$)	J1503 Cost Estimate (\$)	NU Type
J1181 POI-Hazelton 345 Sag Remediation	\$500,000	\$17,831	NRIS
Double Circuit Raun to Sioux City 345 kV	\$35,200,000	\$670,519	SPP
Rebuild S3451 to S3454 345 kV	\$38,481,000	\$2,327,065	SPP
Rebuild S3451 to S3459 345 kV	\$23,418,800	\$1,322,070	SPP
Rebuild S3456 to S3459 345 kV	\$15,444,600	\$860,635	SPP
Columbus 200 MVar Reactive Support	\$3,788,600	\$236,225	SPP
Gentry-Fairport 161 kV Reconductor	\$2,812,100	\$157,924	AECI
Nodaway-Maryville 161 kV Rebuild	\$4,746,600	\$269,930	AECI
Cottleville-Kisker 161 kV Rebuild	\$1,040,000	\$43,522	AECI
Total Cost Per Project:		\$5,905,721	
J1504			
Network Upgrade	NU Cost Estimate (\$)	J1504 Cost Estimate (\$)	NU Type
J1181 POI-Hazelton 345 Sag Remediation	\$500,000	\$17,831	NRIS
Double Circuit Raun to Sioux City 345 kV	\$35,200,000	\$670,519	SPP
Rebuild S3451 to S3454 345 kV	\$38,481,000	\$2,327,065	SPP
Rebuild S3451 to S3459 345 kV	\$23,418,800	\$1,322,070	SPP
Rebuild S3456 to S3459 345 kV	\$15,444,600	\$860,635	SPP
Columbus 200 MVar Reactive Support	\$3,788,600	\$236,225	SPP
Gentry-Fairport 161 kV Reconductor	\$2,812,100	\$157,924	AECI
Nodaway-Maryville 161 kV Rebuild	\$4,746,600	\$269,930	AECI
Cottleville-Kisker 161 kV Rebuild	\$1,040,000	\$43,522	AECI
Total Cost Per Project:		\$5,905,721	



J1520			
Network Upgrade	NU Cost Estimate (\$)	J1520 Cost Estimate (\$)	NU Type
J1181 POI-Hazelton 345 Sag Remediation	\$500,000	\$14,603	NRIS
Gentry-Fairport 161 kV Reconductor	\$2,812,100	\$142,387	AECI
Nodaway-Maryville 161 kV Rebuild	\$4,746,600	\$239,863	AECI
Total Cost Per Project:		\$396,853	
J1528			
Network Upgrade	NU Cost Estimate (\$)	J1528 Cost Estimate (\$)	NU Type
Total Cost Per Project:		\$0	
J1529			
Network Upgrade	NU Cost Estimate (\$)	J1529 Cost Estimate (\$)	NU Type
Total Cost Per Project:		\$0	
J1530			
Network Upgrade	NU Cost Estimate (\$)	J1530 Cost Estimate (\$)	NU Type
Gentry-Fairport 161 kV Reconductor	\$2,812,100	\$286,199	AECI
Nodaway-Maryville 161 kV Rebuild	\$4,746,600	\$499,169	AECI
Total Cost Per Project:		\$785,368	
J1532			
Network Upgrade	NU Cost Estimate (\$)	J1532 Cost Estimate (\$)	NU Type
Gentry-Fairport 161 kV Reconductor	\$2,812,100	\$378,909	AECI
Nodaway-Maryville 161 kV Rebuild	\$4,746,600	\$661,862	AECI
Total Cost Per Project:		\$1,040,771	
J1534			
Network Upgrade	NU Cost Estimate (\$)	J1534 Cost Estimate (\$)	NU Type
J1181 POI-Hazelton 345 Sag Remediation	\$500,000	\$36,642	NRIS



Double Circuit Raun to Sioux City 345 kV	\$35,200,000	\$790,586	SPP
Double Circuit Split Rock to Sioux City 345 kV	\$126,000,000	\$3,200,510	SPP
Double Circuit Raun to S3451 345 kV	\$115,554,682	\$3,510,576	SPP
Rebuild S3451 to S3454 345 kV	\$38,481,000	\$1,059,260	SPP
Rebuild S3451 to S3459 345 kV	\$23,418,800	\$615,850	SPP
Rebuild S3456 to S3459 345 kV	\$15,444,600	\$404,267	SPP
Total Cost Per Project:		\$9,617,691	
J1566			
Network Upgrade	NU Cost Estimate (\$)	J1566 Cost Estimate (\$)	NU Type
J1181 POI-Hazelton 345 Sag Remediation	\$500,000	\$23,442	NRIS
Rutland-Trumant 161 kV Rebuild	\$3,400,000	\$3,400,000	ERIS
Double Circuit Raun to Sioux City 345 kV	\$35,200,000	\$997,726	SPP
Rebuild S3451 to S3454 345 kV	\$38,481,000	\$2,423,391	SPP
Rebuild S3451 to S3459 345 kV	\$23,418,800	\$1,395,034	SPP
Rebuild S3456 to S3459 345 kV	\$15,444,600	\$910,876	SPP
Columbus 200 MVar Reactive Support	\$3,788,600	\$244,275	SPP
Gentry-Fairport 161 kV Reconductor	\$2,812,100	\$155,907	AECI
Cottleville-Kisker 161 kV Rebuild	\$1,040,000	\$43,522	AECI
Nodaway-Maryville 161 kV Rebuild	\$4,746,600	\$264,391	AECI
Double Circuit Split Rock to Sioux City 345 kV	\$126,000,000	\$5,024,057	SPP
Double Circuit Raun to S3451 345 kV	\$115,554,682	\$8,019,770	SPP
Total Cost Per Project:		\$22,902,391	
J1572			
Network Upgrade	NU Cost Estimate (\$)	J1572 Cost Estimate (\$)	NU Type
J1181 POI-Hazelton 345 Sag Remediation	\$500,000	\$25,132	NRIS
Gentry-Fairport 161 kV Reconductor	\$2,812,100	\$154,092	AECI
Nodaway-Maryville 161 kV Rebuild	\$4,746,600	\$261,205	AECI
Cottleville-Kisker 161 kV Rebuild	\$1,040,000	\$42,718	AECI
Double Circuit Raun to Sioux City 345 kV	\$35,200,000	\$817,035	SPP
Double Circuit Split Rock to Sioux City 345 kV	\$126,000,000	\$4,421,498	SPP
Double Circuit Raun to S3451 345 kV	\$115,554,682	\$7,447,200	SPP



Rebuild S3451 to S3454 345 kV	\$38,481,000	\$2,260,305	SPP
Rebuild S3451 to S3459 345 kV	\$23,418,800	\$1,289,492	SPP
Rebuild S3456 to S3459 345 kV	\$15,444,600	\$840,858	SPP
Total Cost Per Project:		\$17,559,535	
J1575			
Network Upgrade	NU Cost Estimate (\$)	J1575 Cost Estimate (\$)	NU Type
J1575 POI-Donaldson 115 Structure Remediation	\$600,000	\$600,000	ERIS
J1181 POI-Hazelton 345 Sag Remediation	\$500,000	\$25,132	NRIS
Donaldson-J1575 Reconductor (OTP LPC)	\$1,000,000	\$1,000,000	OTP LPC
Two Phase Shifters Letellier Substation	\$14,354,026	\$4,739,397	MH
Sheyenne-Maple River 230 kV Jumper and Switch Replacement	\$500,000	\$31,352	MPC
Winger-Wilton 230 kV Rebuild	\$37,000,000	\$6,231,593	MPC
Nary-Helga 115 kV Jumper Replacement	\$100,000	\$16,261	MPC
Drayton-Letellier 230 kV Terminal Upgrade	\$500,000	\$150,960	MPC
Prairie-Walle 230 kV Terminal Upgrade and Structure Raise	\$500,000	\$58,667	MPC
Audubon 40 Mvar MSC +/-100 Mvar STATCOM	\$21,000,000	\$1,350,000	MPC
Wahpeton 230 kV +/-100 Mvar STATCOM	\$20,000,000	\$1,302,409	MPC
Wilton 230 kV 30 Mvar MSC	\$1,000,000	\$65,512	MPC
Winger Two Step 30 Mvar MSC +/-125 Mvar STATCOM	\$24,500,000	\$856,504	MPC
Mahnomn-Ulrich 115 kV Jumper Replacement	\$100,000	\$100,000	MPC
Oslo TN-Oslo 115 kV Jumper Replacement	\$100,000	\$100,000	MPC
Grand Forks-Mill Road 69 kV Sag Mitigation	\$500,000	\$500,000	MPC
Wilton Tap-Scribner 115 kV Jumper Replacement	\$100,000	\$100,000	MPC
Ulrich-Audubon 115 kV Jumper Replacement	\$100,000	\$100,000	MPC
Raun-Sioux City 345 kV Double Circuit	\$35,200,000	\$1,158,331	SPP
Split Rock-Sioux City 345 kV Double Circuit	\$126,000,000	\$3,653,222	SPP
Raun-S3451 Double 345 kV Double Circuit	\$115,554,682	\$3,305,109	SPP
S3451-S3454 345 kV Rebuild	\$38,481,000	\$929,745	SPP
S3451-S3459 345 kV Rebuild	\$23,418,800	\$614,675	SPP
S3456-S3459 345 kV Rebuild	\$15,444,600	\$410,710	SPP
Leeds-Penn Tap 115 kV Rebuild	\$14,290,000	\$2,333,987	SPP



Penn Tap-Devil's Lake 115 kV Rebuild	\$11,400,000	\$1,861,963	SPP
Pleasant Lake-Leeds 115 kV Rebuild	\$16,400,000	\$2,588,619	SPP
Pleasant Lake-Rugby 115 kV Rebuild	\$4,300,000	\$678,723	SPP
White Swan 345 kV 250 Mvar Reactive Support	\$6,222,750	\$305,520	SPP
Columbus 230 kV 200 Mvar Reactive Support	\$3,788,600	\$181,874	SPP
Total Cost Per Project:		\$35,350,265	
J1581			
Network Upgrade	NU Cost Estimate (\$)	J1581 Cost Estimate (\$)	NU Type
J1181 POI-Hazelton 345 Sag Remediation	\$500,000	\$26,791	NRIS
Gentry-Fairport 161 kV Reconductor	\$2,812,100	\$205,724	AECI
Nodaway-Maryville 161 kV Rebuild	\$4,746,600	\$348,746	AECI
Wright City-New Melle 161 kV Reconductor Terminal Upgrade	\$2,413,500	\$148,191	AECI
Wright City 161 kV Bus Upgrades	\$250,000	\$15350	AECI
Double Circuit Raun to Sioux City 345 kV	\$35,200,000	\$3,992,998	SPP
Double Circuit Split Rock to Sioux City 345 kV	\$126,000,000	\$18,329,597	SPP
Double Circuit Raun to S3451 345 kV	\$115,554,682	\$12,878,585	SPP
Rebuild S3451 to S3454 345 kV	\$38,481,000	\$3,800,237	SPP
Rebuild S3451 to S3459 345 kV	\$23,418,800	\$2,281,026	SPP
Rebuild S3456 to S3459 345 kV	\$15,444,600	\$1,495,851	SPP
White Swan 250 MVar Reactive Support	\$6,222,750	\$1,264,014	SPP
Columbus 200 MVar Reactive Support	\$3,788,600	\$369,374	SPP
Total Cost Per Project:		\$45,156,484	
J1582			
Network Upgrade	NU Cost Estimate (\$)	J1582 Cost Estimate (\$)	NU Type
Pine Lake Transformer Upgrade	\$2,500,000	\$2,500,000	NRIS
Three Lakes 115/69 kV Transformer Upgrade	\$2,500,000	\$2,500,000	NRIS
J1181 POI-Hazelton 345 Sag Remediation	\$500,000	\$31,030	NRIS
Double Circuit Raun to Sioux City 345 kV	\$35,200,000	\$1,555,300	SPP
Double Circuit Split Rock to Sioux City 345 kV	\$126,000,000	\$5,971,950	SPP
Double Circuit Raun to S3451 345 kV	\$115,554,682	\$5,760,148	SPP
Rebuild S3451 to S3454 345 kV	\$38,481,000	\$1,708,549	SPP



Rebuild S3451 to S3459 345 kV	\$23,418,800	\$1,025,108	SPP
Rebuild S3456 to S3459 345 kV	\$15,444,600	\$675,408	SPP
Total Cost Per Project:		\$21,727,493	
J1588			
Network Upgrade	NU Cost Estimate (\$)	J1588 Cost Estimate (\$)	NU Type
J1181 POI-Hazelton 345 Sag Remediation	\$500,000	\$35,058	NRIS
Sheyenne Lake Park 230 kV Structure Replacements	\$100,000	\$21,382	NRIS
Two Phase Shifters Letellier Substation	\$14,354,026	\$2,385,699	MH
Sheyenne-Maple River 230 kV Jumper and Switch Replacement	\$500,000	\$392,424	MPC
Winger-Wilton 230 kV Rebuild	\$37,000,000	\$14,664,366	MPC
Nary-Helga 115 kV Jumper Replacement	\$100,000	\$37,566	MPC
Drayton-Letellier 230 kV Terminal Upgrade	\$500,000	\$147,042	MPC
Maple River-Winger 230 kV Sag Remediation	\$500,000	\$500,000	MPC
Prairie-Walle 230 kV Terminal Upgrade and Structure Raise	\$500,000	\$230,631	MPC
Prairie-Lake Ardoch 230 kV Line Rebuild	\$24,000,000	\$11,392,996	MPC
Winger-Walle 230 kV Terminal Upgrade and Structure Raise	\$500,000	\$201,593	MPC
Audubon 40 Mvar MSC +/-100 Mvar STATCOM	\$21,000,000	\$8,450,000	MPC
Edinburg Tap 115 kV 10 Mvar MSC	\$500,000	\$54,783	MPC
Wahpeton 230 kV +/-100 Mvar STATCOM	\$20,000,000	\$7,511,151	MPC
Wilton 230 kV 30 Mvar MSC	\$1,000,000	\$408,133	MPC
Winger Two Step 30 Mvar MSC +/-125 Mvar STATCOM	\$24,500,000	\$9,879,675	MPC
Double Circuit Raun to Sioux City 345 kV	\$35,200,000	\$3,482,351	SPP
Double Circuit Split Rock to Sioux City 345 kV	\$126,000,000	\$11,026,655	SPP
Double Circuit Raun to S3451 345 kV	\$115,554,682	\$9,179,687	SPP
Rebuild S3451 to S3454 345 kV	\$38,481,000	\$2,621,445	SPP
Rebuild S3451 to S3459 345 kV	\$23,418,800	\$1,693,737	SPP
Rebuild S3456 to S3459 345 kV	\$15,444,600	\$1,128,162	SPP
White Swan 250 MVar Reactive Support	\$6,222,750	\$872,186	SPP
Columbus 200 MVar Reactive Support	\$3,788,600	\$525,192	SPP
Total Cost Per Project:		\$86,841,914	
J1590			

Network Upgrade	NU Cost Estimate (\$)	J1590 Cost Estimate (\$)	NU Type
Raun-Remson 345 kV Structure Replacements	\$1,600,000	\$1,600,000	ERIS
J1181 POI-Hazelton 345 Sag Remediation	\$500,000	\$3,892	NRIS
Gentry-Fairport 161 kV Reconductor	\$2,812,100	\$215,943	AECI
Nodaway-Maryville 161 kV Rebuild	\$4,746,600	\$366,679	AECI
Cottleville-Kisker 161 kV Rebuild	\$1,040,000	\$66,680	AECI
Wright City-New Melle 161 kV Reconductor Terminal Upgrade	\$2,413,500	\$164,878	AECI
Wright City 161 kV Bus Upgrades	\$250,000	\$17,079	AECI
Total Cost Per Project:		\$2,435,151	
J1605			
Network Upgrade	NU Cost Estimate (\$)	J1605 Cost Estimate (\$)	NU Type
J1181 POI-Hazelton 345 Sag Remediation	\$500,000	\$41,635	NRIS
Double Circuit Raun to Sioux City 345 kV	\$35,200,000	\$2,427,014	SPP
Double Circuit Split Rock to Sioux City 345 kV	\$126,000,000	\$9,139,624	SPP
Double Circuit Raun to S3451 345 kV	\$115,554,682	\$8,345,323	SPP
Rebuild S3451 to S3454 345 kV	\$38,481,000	\$2,457,406	SPP
Rebuild S3451 to S3459 345 kV	\$23,418,800	\$1,493,808	SPP
Rebuild S3456 to S3459 345 kV	\$15,444,600	\$985,680	SPP
White Swan 250 MVar Reactive Support	\$6,222,750	\$544,297	SPP
Columbus 200 MVar Reactive Support	\$3,788,600	\$340,505	SPP
Total Cost Per Project:		\$25,775,292	
J1611			
Network Upgrade	NU Cost Estimate (\$)	J1611 Cost Estimate (\$)	NU Type
J1181 POI-Hazelton 345 Sag Remediation	\$500,000	\$23,744	NRIS
Double Circuit Raun to Sioux City 345 kV	\$35,200,000	\$1,632,499	SPP
Double Circuit Split Rock to Sioux City 345 kV	\$126,000,000	\$5,873,877	SPP
Double Circuit Raun to S3451 345 kV	\$115,554,682	\$5,140,962	SPP
Rebuild S3451 to S3454 345 kV	\$38,481,000	\$1,501,911	SPP
Rebuild S3451 to S3459 345 kV	\$23,418,800	\$927,524	SPP
Rebuild S3456 to S3459 345 kV	\$15,444,600	\$613,482	SPP



White Swan 250 MVar Reactive Support	\$6,222,750	\$367,018	SPP
Columbus 200 MVar Reactive Support	\$3,788,600	\$230,729	SPP
Total Cost Per Project:		\$16,311,746	
J1620			
Network Upgrade	NU Cost Estimate (\$)	J1620 Cost Estimate (\$)	NU Type
Double Circuit Raun to Sioux City 345 kV	\$35,200,000	\$3,434,102	SPP
Double Circuit Split Rock to Sioux City 345 kV	\$126,000,000	\$13,686,176	SPP
Double Circuit Raun to S3451 345 kV	\$115,554,682	\$7,833,769	SPP
Rebuild S3451 to S3454 345 kV	\$38,481,000	\$2,280,969	SPP
Rebuild S3451 to S3459 345 kV	\$23,418,800	\$1,413,369	SPP
Rebuild S3456 to S3459 345 kV	\$15,444,600	\$933,445	SPP
White Swan 250 MVar Reactive Support	\$6,222,750	\$666,253	SPP
Columbus 200 MVar Reactive Support	\$3,788,600	\$297,017	SPP
Gentry-Fairport 161 kV Reconductor	\$2,812,100	\$122,023	AECI
Nodaway-Maryville 161 kV Rebuild	\$4,746,600	\$206,463	AECI
Total Cost Per Project:		\$30,873,586	
J1622			
Network Upgrade	NU Cost Estimate (\$)	J1622 Cost Estimate (\$)	NU Type
J1181 POI-Hazelton 345 Sag Remediation	\$500,000	\$15,643	NRIS
Sheyenne Lake Park 230 kV Structure Replacements	\$100,000	\$26,260	NRIS
Harvey-J1456 POI 230 kV Structure Replacements	\$2,000,000	\$2,000,000	NRIS
Underwood-J1456 POI 230 kV Structure Replacements	\$1,250,000	\$1,250,000	NRIS
Harvey – Balta 230 kV Structure Replacements	\$750,000	\$750,000	NRIS
Two Phase Shifters Letellier Substation	\$14,354,026	\$2,494,140	MH
Sheyenne-Maple River 230 kV Jumper and Switch Replacement	\$500,000	\$27,122	MPC
Winger-Wilton 230 kV Rebuild	\$37,000,000	\$5,267,519	MPC
Nary-Helga 115 kV Jumper Replacement	\$100,000	\$15,733	MPC
Drayton-Letellier 230 kV Terminal Upgrade	\$500,000	\$69,536	MPC
Prairie-Walle 230 kV Terminal Upgrade and Structure Raise	\$500,000	\$72,855	MPC
Prairie-Lake Ardoch 230 kV Line Rebuild	\$24,000,000	\$4,313,265	MPC
Winger-Walle 230 kV Terminal Upgrade and Structure Raise	\$500,000	\$102,787	MPC



Audubon 40 Mvar MSC +/-100 Mvar STATCOM	\$21,000,000	\$3,975,000	MPC
Edinburg Tap 115 kV 10 Mvar MSC	\$500,000	\$194,783	MPC
Wahpeton 230 kV +/-100 Mvar STATCOM	\$20,000,000	\$3,942,908	MPC
Wilton 230 kV 30 Mvar MSC	\$1,000,000	\$186,747	MPC
Winger Two Step 30 Mvar MSC +/-125 Mvar STATCOM	\$24,500,000	\$4,900,000	MPC
Raun-Sioux City 345 kV Double Circuit	\$35,200,000	\$1,980,525	SPP
Columbus-Meadow Grove 230 kV Double Circuit	\$66,192,123	\$21,664,836	SPP
Split Rock-Sioux City 345 kV Double Circuit	\$126,000,000	\$5,363,372	SPP
Raun-S3451 Double 345 kV Double Circuit	\$115,554,682	\$4,709,762	SPP
S3451-S3454 345 kV Rebuild	\$38,481,000	\$1,313,266	SPP
S3451-S3459 345 kV Rebuild	\$23,418,800	\$889,115	SPP
S3456-S3459 345 kV Rebuild	\$15,444,600	\$596,512	SPP
Leeds-Penn Tap 115 kV Rebuild	\$14,290,000	\$3,821,865	SPP
Penn Tap-Devil's Lake 115 kV Rebuild	\$11,400,000	\$3,048,934	SPP
Pleasant Lake-Leeds 115 kV Rebuild	\$16,400,000	\$4,509,507	SPP
Pleasant Lake-Rugby 115 kV Rebuild	\$4,300,000	\$1,182,371	SPP
White Swan 345 kV 250 Mvar Reactive Support	\$6,222,750	\$641,935	SPP
Columbus 230 kV 200 Mvar Reactive Support	\$3,788,600	\$334,768	SPP
Total Cost Per Project:		\$79,661,066	
J1657			
Network Upgrade	NU Cost Estimate (\$)	J1657 Cost Estimate (\$)	NU Type
Double Circuit Raun to Sioux City 345 kV	\$35,200,000	\$923,878	SPP
Double Circuit Split Rock to Sioux City 345 kV	\$126,000,000	\$3,548,582	SPP
Double Circuit Raun to S3451 345 kV	\$115,554,682	\$3,464,128	SPP
Rebuild S3451 to S3454 345 kV	\$38,481,000	\$1,028,741	SPP
Rebuild S3451 to S3459 345 kV	\$23,418,800	\$615,850	SPP
Rebuild S3456 to S3459 345 kV	\$15,444,600	\$405,755	SPP
Total Cost Per Project:		\$9,986,934	
J1730			
Network Upgrade	NU Cost Estimate (\$)	J1730 Cost Estimate (\$)	NU Type
J1181 POI-Hazelton 345 Sag Remediation	\$500,000	\$49,706	NRIS



Gentry-Fairport 161 kV Reconductor	\$2,812,100	\$237,312	AECI
Nodaway-Maryville 161 kV Rebuild	\$4,746,600	\$397,117	AECI
Cottleville-Kisker 161 kV Rebuild	\$1,040,000	\$68,900	AECI
Wright City-New Melle 161 kV Reconductor Terminal Upgrade	\$2,413,500	\$159,713	AECI
Wright City 161 kV Bus Upgrades	\$250,000	\$16,544	AECI
Double Circuit Raun to Sioux City 345 kV	\$35,200,000	\$3,185,652	SPP
Double Circuit Split Rock to Sioux City 345 kV	\$126,000,000	\$12,166,098	SPP
Double Circuit Raun to S3451 345 kV	\$115,554,682	\$11,486,854	SPP
Rebuild S3451 to S3454 345 kV	\$38,481,000	\$3,397,134	SPP
Rebuild S3451 to S3459 345 kV	\$23,418,800	\$2,048,371	SPP
Rebuild S3456 to S3459 345 kV	\$15,444,600	\$1,350,285	SPP
Total Cost Per Project:		\$34,563,686	
J1734			
Network Upgrade	NU Cost Estimate (\$)	J1734 Cost Estimate (\$)	NU Type
Rebuild Amoco-Madison East	\$180,000	\$180,000	ERIS
Henry NE-Denmark 161 kV Sag Remediation	\$120,000	\$120,000	ERIS
Denmark-Burlington 161 kV Line Rebuild	\$19,700,000	\$19,700,000	ERIS
J1701-Ipava 138 kV Line Rebuild	\$6,000,000	\$791,010	NRIS
Cottleville-Kisker 161 kV Rebuild	\$1,040,000	\$47,885	AECI
Sawyer-Lee 69 kV Rebuild	\$4,139,000	\$3,572,018	AECI
Second 161/69 kV Transformer Lee Substation	\$3,468,000	\$3,042,189	AECI
Total Cost Per Project:		\$27,453,102	
J1754			
Network Upgrade	NU Cost Estimate (\$)	J1754 Cost Estimate (\$)	NU Type
New Magic City-Mallard 115 kV Line	\$4,000,000	\$4,000,000	ERIS
McHenry 230/115 kV Transformer Replacement	\$6,057,149	\$6,057,149	ERIS
J1181 POI-Hazelton 345 Sag Remediation	\$500,000	\$31,045	NRIS
Sheyenne Lake Park 230 kV Structure Replacements	\$100,000	\$52,358	NRIS
Two Phase Shifters Letellier Substation	\$14,354,026	\$4,734,791	MH
Sheyenne-Maple River 230 kV Jumper and Switch Replacement	\$500,000	\$49,102	MPC
Winger-Wilton 230 kV Rebuild	\$37,000,000	\$10,836,522	MPC



Nary-Helga 115 kV Jumper Replacement	\$100,000	\$30,440	MPC
Drayton-Letellier 230 kV Terminal Upgrade	\$500,000	\$132,462	MPC
Prairie-Walle 230 kV Terminal Upgrade and Structure Raise	\$500,000	\$137,847	MPC
Prairie-Lake Ardoch 230 kV Line Rebuild	\$24,000,000	\$8,293,739	MPC
Winger-Walle 230 kV Terminal Upgrade and Structure Raise	\$500,000	\$195,620	MPC
Audubon 40 Mvar MSC +/-100 Mvar STATCOM	\$21,000,000	\$7,225,000	MPC
Edinburg Tap 115 kV 10 Mvar MSC	\$500,000	\$250,435	MPC
Wahpeton 230 kV +/-100 Mvar STATCOM	\$20,000,000	\$7,243,533	MPC
Wilton 230 kV 30 Mvar MSC	\$1,000,000	\$339,608	MPC
Winger Two Step 30 Mvar MSC +/-125 Mvar STATCOM	\$24,500,000	\$8,863,821	MPC
Gentry-Fairport 161 kV Reconductor	\$2,812,100	\$152,747	AECI
Nodaway-Maryville 161 kV Rebuild	\$4,746,600	\$255,778	AECI
Cottleville-Kisker 161 kV Rebuild	\$1,040,000	\$57,427	AECI
Wright City-New Melle 161 kV Reconductor Terminal Upgrade	\$2,413,500	\$140,643	AECI
Wright City 161 kV Bus Upgrades	\$250,000	\$14,568	AECI
Raun-Sioux City 345 kV Double Circuit	\$35,200,000	\$4,019,709	SPP
Columbus-Meadow Grove 230 kV Double Circuit	\$66,192,123	\$44,527,287	SPP
Split Rock-Sioux City 345 kV Double Circuit	\$126,000,000	\$10,939,090	SPP
Raun-S3451 Double 345 kV Double Circuit	\$115,554,682	\$10,036,007	SPP
S3451-S3454 345 kV Rebuild	\$38,481,000	\$2,758,145	SPP
S3451-S3459 345 kV Rebuild	\$23,418,800	\$1,908,243	SPP
S3456-S3459 345 kV Rebuild	\$15,444,600	\$1,283,616	SPP
Leeds-Penn Tap 115 kV Rebuild	\$14,290,000	\$8,134,147	SPP
Penn Tap-Devil's Lake 115 kV Rebuild	\$11,400,000	\$6,489,103	SPP
Pleasant Lake-Leeds 115 kV Rebuild	\$16,400,000	\$9,301,873	SPP
Pleasant Lake-Rugby 115 kV Rebuild	\$4,300,000	\$2,438,906	SPP
White Swan 345 kV 250 Mvar Reactive Support	\$6,222,750	\$1,320,484	SPP
Columbus 230 kV 200 Mvar Reactive Support	\$3,788,600	\$688,042	SPP
Total Cost Per Project:		\$162,939,287	
J1769			
Network Upgrade	NU Cost Estimate (\$)	J1769 Cost Estimate (\$)	NU Type



Raun-S3451 Double 345 kV Double Circuit	\$115,554,682	\$2,611,186	SPP
S3451-S3454 345 kV Rebuild	\$38,481,000	\$785,861	SPP
S3451-S3459 345 kV Rebuild	\$23,418,800	\$457,574	SPP
S3456-S3459 345 kV Rebuild	\$15,444,600	\$301,042	SPP
Total Cost Per Project:		\$4,155,663	
J1804			
Network Upgrade	NU Cost Estimate (\$)	J1804 Cost Estimate (\$)	NU Type
Total Cost Per Project:		0	
J1808			
Network Upgrade	NU Cost Estimate (\$)	J1808 Cost Estimate (\$)	NU Type
Double Circuit Raun to Sioux City 345 kV	\$35,200,000	\$1,306,492	SPP
Double Circuit Split Rock to Sioux City 345 kV	\$126,000,000	\$4,984,608	SPP
Double Circuit Raun to S3451 345 kV	\$115,554,682	\$4,864,440	SPP
Rebuild S3451 to S3454 345 kV	\$38,481,000	\$1,443,963	SPP
Rebuild S3451 to S3459 345 kV	\$23,418,800	\$865,445	SPP
Rebuild S3456 to S3459 345 kV	\$15,444,600	\$570,461	SPP
Total Cost Per Project:		\$14,035,409	
J1826			
Network Upgrade	NU Cost Estimate (\$)	J1826 Cost Estimate (\$)	NU Type
Double Circuit Raun to Sioux City 345 kV	\$35,200,000	\$41,470	SPP
Double Circuit Split Rock to Sioux City 345 kV	\$126,000,000	\$164,666	SPP
Double Circuit Raun to S3451 345 kV	\$115,554,682	\$162,549	SPP
Rebuild S3451 to S3454 345 kV	\$38,481,000	\$48,334	SPP
Rebuild S3451 to S3459 345 kV	\$23,418,800	\$28,823	SPP
Rebuild S3456 to S3459 345 kV	\$15,444,600	\$18,965	SPP
Total Cost Per Project:		\$464,807	

1.3.

Total Network Upgrades



The cost allocation of Network Upgrades for the projects in the DPP 2020 West Phase 2 reflects responsibilities for mitigating system impacts. The total cost of network upgrades is listed in Table 1 below. The costs for Network Upgrades are planning-level estimates and subject to revision in the facility studies.

Table 1: Total Cost of Network Upgrades for DPP 2020 West Phase 2 Projects

Project	ERIS Network Upgrades (\$)					NRIS Network Upgrades (\$)	Interconnection Facilities (\$)		Shared Network Upgrades (\$)	Total Network Upgrade Cost for Milestone Calculation (\$)	M2 (\$)	M3 (\$)	M4 (\$)
	Steady State	LPC Studies	Stability	Short Circuit	Affected System	Deliverability	TO Network Upgrades	TO – Owned Direct Assigned					
A	B	C	D	E	F	G	H	I	J	k = b+c+d+e+g+h+j	\$4,000/MW at Application	10% of (k) from Phase I-M2	20% of (k) from Phase 2-M3-M2
J1494	\$0	\$0	\$0	\$0	\$5,827,662	\$10,122	\$3,099,000	\$676,000	\$0	\$3,109,122	\$200,000	\$152,074	\$269,750
J1495	\$0	\$0	\$0	\$0	\$4,950,934	\$15,402	\$1,602,000	\$1,258,000	\$0	\$1,617,402	\$200,000	\$203,153	(\$79,673)
J1498	\$0	\$0	\$0	\$0	\$11,002,936	\$10,007,437	\$3,712,000	\$1,134,000	\$0	\$13,719,437	\$200,000	\$854,200	\$1,689,687
J1503	\$0	\$0	\$0	\$0	\$5,885,891	\$17,831	\$4,500,000	\$0	\$0	\$4,517,831	\$600,000	\$0	\$303,566
J1504	\$0	\$0	\$0	\$0	\$5,885,891	\$17,831	\$4,500,000	\$0	\$0	\$4,517,831	\$600,000	\$0	\$303,566
J1520	\$0	\$0	\$0	\$0	\$382,250	\$14,603	\$10,000,000	\$0	\$0	\$10,014,603	\$540,000	\$467,780	\$995,141
J1528	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0
J1529	\$0	\$0	\$0	\$0	\$0	\$0	\$12,000,000	\$0	\$0	\$12,000,000	\$200,000	\$1,620,425	\$579,575
J1530	\$0	\$0	\$0	\$0	\$785,368	\$0	\$10,000,000	\$0	\$0	\$10,000,000	\$200,000	\$849,181	\$950,819
J1532	\$0	\$0	\$0	\$0	\$1,040,771	\$0	\$12,000,000	\$0	\$0	\$12,000,000	\$200,000	\$1,120,475	\$1,079,525
J1534	\$0	\$0	\$0	\$0	\$9,581,049	\$36,642	\$0	\$0	\$0	\$36,642	\$400,000	\$0	(\$392,672)
J1566	\$3,400,000	\$0	\$0	\$0	\$19,478,949	\$23,442	\$2,500,000	\$0	\$0	\$5,923,442	\$600,000	\$0	\$584,688
J1572	\$0	\$0	\$0	\$0	\$17,534,404	\$25,132	\$10,168,000	\$1,176,000	\$0	\$10,193,132	\$600,000	\$435,583	\$1,003,043
J1575	\$600,000	\$1,000,000	\$0	\$0	\$33,725,136	\$12,234	\$4,000,000	\$0	\$0	\$5,612,234	\$280,000	\$651,969	\$190,478
J1581	\$0	\$0	\$0	\$0	\$45,129,694	\$26,791	\$3,562,000	\$873,000	\$0	\$3,588,791	\$800,000	\$0	(\$82,242)
J1582	\$0	\$0	\$0	\$0	\$16,696,464	\$5,031,030	\$8,927,000	\$1,168,000	\$0	\$13,958,030	\$640,000	\$718,900	\$1,432,706
J1588	\$0	\$0	\$0	\$0	\$86,785,474	\$56,440	\$4,079,000	\$2,043,000	\$0	\$4,135,440	\$800,000	\$564,262	(\$537,174)
J1590	\$1,600,000	\$0	\$0	\$0	\$831,259	\$3,892	\$2,500,000	\$0	\$0	\$4,103,892	\$800,000	\$0	\$20,778
J1605	\$0	\$0	\$0	\$0	\$25,733,657	\$41,635	\$2,910,000	\$1,851,000	\$0	\$2,951,635	\$800,000	\$0	(\$209,673)
J1611	\$0	\$0	\$0	\$0	\$16,288,002	\$23,744	\$8,317,486	\$400,544	\$0	\$8,341,230	\$696,000	\$2,737,073	(\$1,764,827)
J1620	\$0	\$0	\$0	\$0	\$30,873,585	\$0	\$8,808,000	\$1,164,000	\$0	\$8,808,000	\$500,000	\$872,756	\$388,844
J1622	\$0	\$0	\$0	\$0	\$75,619,163	\$4,041,904	\$0	\$0	\$0	\$4,041,904	\$400,000	\$0	\$408,381

Project	ERIS Network Upgrades (\$)					NRIS Network Upgrades (\$)	Interconnection Facilities (\$)		Shared Network Upgrades (\$)	Total Network Upgrade Cost for Milestone Calculation (\$)	M2 (\$)	M3 (\$)	M4 (\$)
	Steady State	LPC Studies	Stability	Short Circuit	Affected System	Deliverability	TO Network Upgrades	TO – Owned Direct Assigned					
A	B	C	D	E	F	G	H	I	J	k = b+c+d+e+g+h+j	\$4,000/MW at Application	10% of (k) from Phase I-M2	20% of (k) from Phase 2-M3-M2
J1657	\$0	\$0	\$0	\$0	\$9,986,935	\$0	\$0	\$0	\$0	\$0	\$400,000	\$651,310	(\$1,051,310)
J1730	\$0	\$0	\$0	\$0	\$34,513,980	\$49,706	\$14,477,000	\$1,938,000	\$0	\$14,526,706	\$1,200,000	\$112,365	\$1,592,976
J1734	\$20,000,000	\$0	\$0	\$0	\$6,662,092	\$791,010	\$1,250,000	\$0	\$0	\$22,041,010	\$600,000	\$2,285,000	\$1,523,202
J1754	\$10,057,149	\$0	\$0	\$0	\$152,798,735	\$83,403	\$12,488,901	\$1,461,537	\$0	\$22,629,453	\$800,000	\$4,025,958	(\$300,067)
J1769	\$0	\$0	\$0	\$0	\$4,155,663	\$0	\$65,000	\$0	\$0	\$65,000	\$480,000	\$4,079,590	(\$4,546,590)
J1804	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0
J1808	\$0	\$50,000	\$0	\$0	\$14,035,409	\$0	\$623,937	\$814,430	\$0	\$673,937	\$800,000	\$5,176,300	(\$5,841,513)
J1826	\$0	\$0	\$0	\$0	\$464,807	\$0	\$1,556,000	\$1,325,000	\$0	\$1,556,000	\$16,000	\$475,125	(\$179,925)
Total	\$35,657,149	\$1,050,000	\$0	\$0	\$636,656,160	\$26,330,231	\$147,645,324	\$17,282,511	\$0	\$204,682,704			

Analyses performed demonstrate the following transmission facilities are required to reliably interconnect this group of generators to the transmission system. Energy Resource Interconnection Service (ERIS) Network Upgrades and Network Resource Interconnection Service (NRIS) Network Upgrades are shown in Table 2. Shared Network Upgrades are shown in Table 4.

Table 2: ERIS & NRIS Upgrades (Planning level cost estimates)

Network Upgrade	Cost (\$)	TO	Upgrade Type	Self Fund
J1575 POI-Donaldson 115 Structure Remediation	\$600,000	OTP	ERIS	Yes
Raun-Remson 345 kV Structure Remediation	\$1,600,000	MEC	ERIS	Yes
New Magic City-Mallard 115 kV Line	\$4,000,000	XEL	ERIS	Yes
McHenry 230/115 kV Transformer Upgrade	\$6,057,149	GRE	ERIS	No
Rebuild Amoco-Madison East	\$180,000	AMRN	ERIS	Yes
Henry NE-Denmark 161 kV Sag Remediation	\$120,000	MEC	ERIS	Yes
Denmark-Burlington 161 kV Line Rebuild	\$19,700,000	MEC	ERIS	Yes
Rutland-Trumant 161 kV Rebuild	\$3,400,000	ITCM/SMMPA	ERIS	Yes

Network Upgrade	Cost (\$)	TO	Upgrade Type	Self Fund
3 rd Nobles 345/115 kV Transformer	\$10,000,000	XEL	NRIS	Yes
Three Lakes 115/69 kV Transformer Replacement	\$2,500,000	XEL	NRIS	Yes
J1701-Ipava 138 kV Line Rebuild	\$6,000,000	AMRN	NRIS	Yes
Sheyenne Lake Park 230 kV Structure Replacements	\$100,000	OTP	NRIS	Yes
Pine Lake 161/115 kV Transformer Upgrade	\$2,500,000	XEL	NRIS	Yes
Harvey-J1456 POI 230 kV Structure Replacements	\$2,000,000	OTP	NRIS	Yes
Underwood-J1456 POI 230 kV Structure Replacements	\$1,250,000	OTP	NRIS	Yes
J1181 POI-Hazelton Structure Replacements	\$500,000	MEC	NRIS	Yes
Harvey – Balta 230 kV Structure Replacements	\$750,000	OTP	NRIS	Yes
Elk Mound-Wheaton 161 kV Jumper/Disconnect Upgrade (DPC LPC)	\$50,000	DPC	ERIS	Yes
Donaldson-J1575 Reconductor (OTP LPC)	\$1,000,000	OTP	ERIS	Yes
Raun-Sioux City 345 kV Double Circuit	\$35,200,000	WAPA/MEC	AFS	MEC:Yes
Columbus-Meadow Grove 230 kV Double Circuit	\$66,192,123	NPPD	AFS	N/A
Split Rock-Sioux City 345 kV Double Circuit	\$126,000,000	WAPA/XEL	AFS	XEL:Yes
Raun-S3451 Double 345 kV Double Circuit	\$115,554,682	OPPD/MEC	AFS	MEC:Yes
S3451-S3454 345 kV Rebuild	\$38,481,000	OPPD	AFS	N/A
S3451-S3459 345 kV Rebuild	\$23,418,800	OPPD	AFS	N/A
S3456-S3459 345 kV Rebuild	\$15,444,600	OPPD	AFS	N/A
Leeds-Penn Tap 115 kV Rebuild	\$14,290,000	WAPA	AFS	N/A
Penn Tap-Devil's Lake 115 kV Rebuild	\$11,400,000	WAPA	AFS	N/A
Pleasant Lake-Leeds 115 kV Rebuild	\$16,400,000	WAPA	AFS	N/A
Pleasant Lake-Rugby 115 kV Rebuild	\$4,300,000	WAPA	AFS	N/A
White Swan 345 kV 250 Mvar Reactive Support	\$6,222,750	WAPA	AFS	N/A
Columbus 230 kV 200 Mvar Reactive Support	\$3,788,600	NPPD	AFS	N/A
Sheyenne-Maple River 230 kV Jumper and Switch Replacement	\$500,000	MPC	AFS	N/A
Mahnomn-Ulrich 115 kV Jumper Replacement	\$100,000	MPC	AFS	N/A
Oslo TN-Oslo 115 kV Jumper Replacement	\$100,000	MPC	AFS	N/A
Grand Forks-Mill Road 69 kV Sag Mitigation	\$500,000	MPC	AFS	N/A
Wilton Tap-Scribner 115 kV Jumper Replacement	\$100,000	MPC	AFS	N/A

Network Upgrade	Cost (\$)	TO	Upgrade Type	Self Fund
Nary-Helga 115 kV Jumper Replacement	\$100,000	MPC	AFS	N/A
Ulrich-Audubon 115 kV Jumper Replacement	\$100,000	MPC	AFS	N/A
Drayton-Letellier 230 kV Terminal Upgrade	\$500,000	MPC	AFS	N/A
Maple River-Winger 230 kV Sag Remediation	\$500,000	MPC	AFS	N/A
Prairie-Walle 230 kV Terminal Upgrade and Structure Raise	\$500,000	MPC	AFS	N/A
Prairie-Lake Ardoch 230 kV Line Rebuild	\$24,000,000	MPC	AFS	N/A
Winger-Walle 230 kV Terminal Upgrade and Structure Raise	\$500,000	MPC	AFS	N/A
Audubon 40 Mvar MSC +/-100 Mvar STATCOM	\$21,000,000	MPC	AFS	N/A
Edinburg Tap 115 kV 10 Mvar MSC	\$500,000	MPC	AFS	N/A
Wahpeton 230 kV +/-100 Mvar STATCOM	\$20,000,000	MPC	AFS	N/A
Wilton 230 kV 30 Mvar MSC	\$1,000,000	MPC	AFS	N/A
Winger Two Step 30 Mvar MSC +/-125 Mvar STATCOM	\$24,500,000	MPC	AFS	N/A
Winger-Wilton 230 kV Rebuild	\$37,000,000	MPC	AFS	N/A
Gentry-Fairport 161 kV Reconductor	\$2,812,100	AECI	AFS	N/A
Nodaway-Maryville 161 kV Rebuild	\$4,746,600	AECI	AFS	N/A
Cottleville-Kisker 161 kV Rebuild	\$1,040,000	AECI	AFS	N/A
Sawyer-Lee 69 kV Rebuild	\$4,139,000	AECI	AFS	N/A
Wright City-New Melle 161 kV Reconductor Terminal Upgrade	\$2,413,500	AECI	AFS	N/A
Wright City 161 kV Bus Upgrades	\$250,000	AECI	AFS	N/A
Second 161/69 kV Transformer Lee Substation	\$3,468,000	AECI	AFS	N/A
Two Phase Shifters Letellier Substation	\$14,354,026	MH	AFS	N/A

Table 3: Shared Network Upgrades (Planning level cost estimates)

Shared Network Upgrade	TO	Higher queued projects associated with SNU	Study projects associated with SNU	Cost of solution (\$)
None				



Note:

- 1) Details pertaining to upgrades, costs, and the execution plan for interconnection of the generating facility at the POI will be documented in the Facility Study for Interconnecting Generator.
- 2) Facilities that have been included as base case assumptions and the level of interconnection service that would be conditional upon these facilities being in service will be documented in the GIA (Generator Interconnection Agreement) for each respective GI request successfully achieving GIA execution.

2. FERC Order 827 Compliance Review

The Final Rule of FERC Order 827 “Reactive Power Requirements for Non-Synchronous Generation”, which was issued June 16, 2016, stated that “Under this Final Rule, newly interconnecting non-synchronous generators that have not yet executed a Facilities Study Agreement as of the effective date of this Final Rule will be required to provide dynamic reactive power within the range of 0.95 leading to 0.95 lagging at the high-side of the generator substation.” As such, this Final Rule applies to all non-synchronous projects included in the DPP 2020 West study cycle.

In this study, the power factor at the high-side of the generator substation for each project was calculated and reviewed. The study method is to set Qgen of each study project at its Qmax, solve the case, then record the P and Q injection on the high side of the generator substation to calculate the lagging power factor (injecting VAR to the system). The same process is then repeated by setting Qgen at Qmin to calculate the leading power factor (absorbing VAR from the system).

The results show that not all projects meet FERC Order 827 requirements. Additional reactive support will be needed for these projects to meet the FERC requirement on reactive power capability prior to the completion of their GIA.

Table 4: FERC Order 827 Review Results

Project	Pmax (MW)	Reactive Power Capability (MVar)	Proposed VAR Compensation	VAR Injection			VAR Absorption			FERC Order 827 Compliant	MVar Deficiency
				P (MW)	Q (MVar)	Lagging p.f (pu)	P (MW)	Q (MVar)	Leading p.f. (pu)		
J1494	52.8	+/- 38.184	N/A	50	28.3	0.870	49.7	54.8	0.672	Yes	N/A
J1495	52.8	+/- 38.184	N/A	50	28.3	0.870	49.7	54.8	0.672	Yes	N/A
J1498	52.8	+/-20.9	Single Step 7 MVar	50	22.3	0.913	50	27.8	0.874	Yes	N/A
J1503	150	+/-72.9	Single Step 4 MVar	146.1	50	0.946	143.7	121.5	0.764	Yes	N/A
J1504	150	+/-72.9	Single Step 4 MVar	146.1	50	0.946	143.7	121.5	0.764	Yes	N/A
J1520	135	+/-65.4	Single Step 23 MVar	131.8	41.1	0.955	129.9	110.8	0.761	No	2
J1528*	76.6	+/-47.3	Four Steps 6 MVar	75.5	70.4	0.731	75.2	52.9	0.818	Yes	N/A
J1529	50.12	+/-24.22	Two Steps 5 MVar	49.7	19.6	0.930	49.5	31.3	0.845	Yes	N/A
J1530	50.12	+/-24.22	Two Steps 5 MVar	49.7	19.6	0.930	49.5	31.3	0.845	Yes	N/A

Project	Pmax (MW)	Reactive Power Capability (MVar)	Proposed VAR Compensati on	VAR Injection			VAR Absorption			FERC Order 827 Compliant	MVar Deficiency
				P (MW)	Q (MVar)	Lagging p.f (pu)	P (MW)	Q (MVar)	Leading p.f. (pu)		
J1532	50.12	+/-24.22	Two Steps 5 MVar	49.7	19.6	0.930	49.5	31.3	0.845	Yes	N/A
J1534	102	+/-78.2	N/A	100. 3	58	0.866	99	114	0.656	Yes	N/A
J1566	182.4	+/-50	Eight Steps 5 MVar	150. 4	78.6	0.886	150	78.6	0.886	Yes	N/A
J1572	227.15	+/- 139.818	N/A	149. 8	104.3	0.821	146.9	230. 4	0.538	Yes	N/A
J1575	72.8	+34.6/- 25.94	Two Steps 5 MVar	72.2	32.8	0.910	71.9	41	0.869	Yes	N/A
J1581	243.2	+/-66	Six Steps 7 MVar	200. 4	67	0.948	198.6	141. 6	0.814	Yes	N/A
J1582**	165.2	+/-82.38	Two Steps 6 MVar	160	59.7	0.937	157.6	94.6	0.857	Yes	N/A
J1588	243.2	+/-66	Eight Steps 4 MVar	200. 2	60.4	0.957	198.6	122. 6	0.851	No	5
J1590	200	+/-96.9	Two Steps 6 MVar	196. 9	73.3	0.937	196.3	137. 2	0.820	Yes	N/A
J1605	243.2	+/-66	Six Steps 7 MVar	200. 4	67	0.948	198.6	141. 6	0.814	Yes	N/A
J1611	212.8	+/-58	Eight Steps 7 MVar	199. 8	60.6	0.957	198	136	0.824	No	5
J1620	125	+/-41.08	Two Steps 16 MVar	123. 8	64.6	0.887	123.2	66	0.881	Yes	N/A
J1622	102	+/-78.19	N/A	100. 3	58	0.866	99	114	0.656	Yes	N/A
J1657	100.75	+/- 52.482	N/A	98.9	41.7	0.921	97.8	75.8	0.790	Yes	N/A
J1730*	310.1	+/-181.8	Four Steps 6 MVar	297	97.9	0.950	285.8	263	0.736	Yes	N/A

Project	Pmax (MW)	Reactive Power Capability (MVar)	Proposed VAR Compensation	VAR Injection			VAR Absorption			FERC Order 827 Compliant	MVar Deficiency
				P (MW)	Q (MVar)	Lagging p.f (pu)	P (MW)	Q (MVar)	Leading p.f. (pu)		
J1734	150	+/-72.8	N/A	146.1	50	0.946	143.7	121.5	0.764	Yes	N/A
J1754	208	+42.236 /-60.667	N/A	202.3	7.2	0.999	197.9	121.8	0.852	No	59
J1769	120	+/-49.908	One Step 8 MVar	118.3	35.4	0.958	117.4	85	0.810	No	3
J1804**	40.8	+/-25.3	Two Steps 6 MVar	40.4	40.1	0.710	40.4	25.5	0.846	Yes	N/A
J1808	200	+/-101	N/A	194.9	54.7	0.963	192.1	122.4	0.843	No	9
J1826	4.4	+/-4.85	N/A	4	4.3	0.681	3.9	5.7	0.565	Yes	N/A
*Hybrid Interconnection Requests											
**Hybrid Interconnection Requests											

3. Model Development and Study Assumptions

3.1. Base Case Models

The origin of the DPP 2020 West models is based on the MTEP 20 series models with the Bench Cases including all prior-queued projects and their associated network upgrades accounted for.

- Bench Cases
 - DPP20-2025SH90-PhaseII-Bench_Final
 - DPP20-2025SUM-PhaseII-Bench_Final
- Study Cases
 - DPP20-2025SUM-PhaseII-Study_Final
 - DPP20-2025SH90-PhaseII-Study_Final
 - DPP20-2025SH90-PhaseII-Study_Charging_Final

3.2. Monitored Elements

Under NERC category P0 conditions (system intact) branches were monitored for loading above the normal rating (PSS®E Rating A), and for NERC category P1-P7 conditions branches were monitored for emergency rating (PSS®E Rating B). Voltage limits were specified for system intact and contingent conditions as per applicable Transmission Owner Planning Criteria.

3.3. Contingencies

The following contingencies were considered in the steady state analysis:

- 1) NERC Category P0 (system intact -- no contingencies)
- 2) NERC Category P1 contingencies
 - a. Single element outages, at buses with a nominal voltage of 68 kV and above
 - b. Multiple element NERC Category P1 contingencies
- 3) NERC Category P2-P7 contingencies
- 4) For all the contingencies and post-disturbance analyses, cases were solved with transformer tap adjustment enabled, area interchange adjustment disabled, phase shifter adjustment disabled (fixed) and switched shunt adjustment enabled.

3.4. Study Methodology

Non-linear (AC) contingency analysis was performed on the benchmark and study cases, and the incremental impact of the DPP 2020 West generating facilities was evaluated by comparing the steady state performance of the transmission system in the Bench and Study Cases. Analyses used PSS®E version 33.11.0 and TARA version 1801d.

3.5. Performance Criteria

A branch is considered a thermal constraint if the following conditions are met:

- 1) The generator has a larger than twenty percent (20%) sensitivity factor on the overloaded facilities under post-contingent condition (see NERC TPL) or five percent (5%) sensitivity factor under system-intact condition, or
- 2) The overloaded facility or the overload-causing contingency is at generator's outlet, or
- 3) The megawatt impact due to the generator is greater than or equal to twenty percent (20%) of the applicable rating (normal or emergency) of the overloaded facility, or
- 4) For any other constrained facility, where none of the Study Generators meet one of the above criteria, however, the cumulative MW impact of the group of study generators is greater than twenty percent (20%) of the rating of the facility, then only those study generators whose individual MW impact is greater than five percent (5%) of the rating of the facility and has DF greater than five percent (5%) will be responsible for mitigating the cumulative MW impact constraint, or
- 5) Impacts on Affected Systems would be classified as Injection constraints based on the Affected Systems' criteria, or
- 6) Any other applicable Transmission Owner FERC filed Local Planning Criteria are met.

A bus is considered a voltage constraint if both of the following conditions are met:

- 1) The bus voltage is outside of the applicable normal or emergency limits for the post change case, and
- 2) The change in bus voltage is greater than 0.01 per unit

All generators must mitigate thermal injection constraints and voltage constraints in order to obtain any type of Interconnection Service. Further, all generators requesting Network Resource Interconnection Service (NRIS) must mitigate constraints found by using the Deliverability algorithm, to meet the system performance criteria for NERC category P1 events, if DFAX due to the study generator is equal to or greater than 5%.

4. Backbone Network Upgrade Analysis

No Backbone Upgrades were determined to be needed for the DPP 2020 Cycle.

5. Thermal Analysis

The thermal analysis results for 2020 West group show generator projects causing constraints. The details pertaining to the thermal analysis can be found in Appendix A — .

6. Voltage Analysis

The voltage analysis results for summer peak models show that voltage violations are observed in both the peak and off peak scenarios. Results of this analysis are included in Appendix B. A voltage stability analysis was also performed on the MWEX interface. A detailed report is in Appendix D.

7. Stability Analysis

The stability analysis ran a set of MTEP faults along with 3 Phase and SLG faults at the Point of Interconnections for all study units. Plot results from the stability analysis are included in appendix I of this report.

8. Short Circuit Analysis

Short Circuit Analysis was performed for all study units to ensure that sufficient breaker interrupting capabilities are maintained. The results of all short circuit analysis are included in Appendix J.

9. Affected System Impact Study

Southwest Power Pool, PJM, Minnkota Power Cooperative, Manitoba Hydro, and Associative Electric Cooperative Incorporated have flagged West projects as potentially impacting their respective systems. All Affected System Reports are included in appendix E.

10. Deliverability Analysis

10.1. Introduction

Generator interconnection projects have to pass Generator Deliverability Study to be granted NRIS. Any generator that has a 5% DFAX on any constrained flowgate is undeliverable. If the generator is deemed not fully deliverable, the customer, at decision point, can choose either to change the project to an Energy Resource (ER) project or nominate a lower NRIS amount. Generator Deliverability Study ensures that the Network Resources, on an aggregate basis, can meet the MISO aggregate load requirements during system peak condition without getting “bottled up”.

MISO Generator Deliverability Study whitepaper describing the algorithm can be found in BPM 015 – Generation Interconnection, Appendix C.

10.2. Determining the MW Restriction

If one facility is overloaded based on the assessed “severe yet credible dispatch” scenario described in the study methodology, and the generator under study has a DF greater than 5%, part or all of its output is not deliverable. The restricted MW is calculated as following:

$$(\text{MW restricted}) = (\text{worst loading} - \text{MW rating}) / (\text{generator sensitivity factor})$$

If the result is larger than the maximum output of the generator, 100% of this generator’s output is not deliverable.

10.3. Deliverability Study Results

The deliverability analysis assumes all ERIIS upgrades as those upgrades are required for the study group to interconnection to the transmission system. With all the MISO analysis **ERIS upgrades assumed as well as the prior queued network upgrades**, the deliverability analysis determined NRIS service amounts that can be granted with no network upgrades. If a constraint is identified, the deliverability is calculated by taking the NRIS Request amount subtracting the calculated shared deductible. The shared deduction is the amount by which each study generator affecting a given flowgate must be scaled down in order to prevent loading violations on the flowgate. The deduction is proportionally taken from each study generator affecting a particular flowgate. If a particular monitored element becomes a flowgate under multiple contingencies, the contingency with the highest total shared deduction is used. Detailed tables on the breakdown of NRIS amounts with network upgrades are in Appendix F.

J1494 is deliverable to 43.31 MWs without network upgrades
J1495 is deliverable to 43.31 MWs without network upgrades
J1498 is deliverable to 0 MWs without network upgrades
J1503 is deliverable to 103.94 MWs without network upgrades
J1504 is deliverable to 103.94 MWs without network upgrades
J1520 is deliverable to 93.55 MWs without network upgrades
J1528 did not request NRIS
J1529 is fully deliverable to 50 MWs
J1529 is fully deliverable to 50 MWs
J1529 is fully deliverable to 50 MWs

J1534 is deliverable to 86.62 MWs without network upgrades
J1566 is deliverable to 129.92 MWs without network upgrades
J1572 is deliverable to 129.92 MWs without network upgrades
J1575 is deliverable to 60.66 MWs without network upgrades
J1581 is deliverable to 182.24 MWs without network upgrades
J1582 is deliverable to 52.96 MWs without network upgrades
J1588 is deliverable to 173.24 MWs without network upgrades
J1590 is deliverable to 37.32 MWs without network upgrades
J1605 is deliverable to 173.3 MWs without network upgrades
J1611 is deliverable to 103.94 MWs without network upgrades
J1620 did not request NRIS
J1622 is deliverable to 46.82 MWs without network upgrades
J1657 did not request NRIS
J1730 is deliverable to 259.86 MWs without network upgrades
J1734 is deliverable to 0 MWs without network upgrades
J1754 is deliverable to 173.24 MWs without network upgrades
J1766 is deliverable to 0 MWs without network upgrades
J1769 did not request NRIS
J1804 did not request NRIS
J1808 did not request NRIS
J1826 did not request NRIS

11. Shared Network Upgrades Analysis

Shared Network Upgrade (SNU) Analysis tests for Network Upgrades driven by higher queued interconnection projects was performed for this System Impact Study. SNUs were not identified for DPP 2020 West Area requests.

12. Cost Allocation

The cost allocation of Network Upgrades for the study group reflects responsibilities for mitigating system impacts based on Interconnection Customer-elected level of Interconnection service as of the draft System Impact Study report date.

12.1. Cost Assumptions for Network Upgrades

The cost estimate for each network upgrade identified in System Impact Study was provided by the corresponding transmission owning company.

12.2. Cost Allocation Methodology

The costs of Network Upgrades (NU) for a set of generation projects (one or more sub-groups or entire group with identified NU) are allocated based on the MW impact from each project on the constrained facilities in the Study Case.

Cost Allocation Methodology for Thermal Constraints

1. With all Study Group generation projects dispatched in the Study Case, all thermal constraints are identified.
2. Distribution factor from each project on each constraint is obtained.
3. For each thermal constraint, the maximum MW contribution (increasing flow) from each project is then calculated in the Post Case without any network upgrades.
4. For each thermal constraint, the cost estimates for one or a subset of NU are provided by the corresponding Transmission Owner.
5. Then the cost of each NU is allocated based on the pro rata share of the MW contribution from each project on the constraints mitigated or partly mitigated by this NU. The methodology to determine the cost allocation of one NU is:

$$\text{Cost of NU} = \frac{\text{Project A cost portion of NU} \times \text{Max(Proj. A MW contribution on constraint)}}{\sum_i \text{Max(Proj. i MW contrution on constraint)}}$$

6. The total NU costs for each project are calculated if more than one NU is required.

Cost Allocation Methodology for Voltage Constraints

Cost allocation of voltage constraint driven network upgrades will be determined by the pro rata share of the voltage impact each project has on the most constrained bus under the most constraining contingency. The voltage impact of each project will be calculated by locking all voltage regulating equipment in the model and then backing out each GI project one at a time to identify each project's impact to the constraint. In severe instances of voltage collapse where projects cannot be backed out one at a time, they will be added one at a time to determine their impact to the constraint.

As the number and types of constraints increases, mitigating the constraints individually may result in higher overall costs. In instances when mitigation(s) resolve multiple types of constraints (such as thermal + voltage or thermal + voltage + transient stability) the cost is allocated based off the ratio share of the total cost of the independent mitigations in order to equitably allocate the cost to all parties contributing to constraints. In summary, only the lowest cost mitigation option will be constructed, but for cost allocation purposes the independent mitigations are required.

Appendix A – Thermal Analysis Results

Thermal Constraints

Appendix B – Voltage Constraints

Voltage Constraints

Appendix C – Network Upgrade Cost Allocation

Appendix D – MWEX Voltage Stability Report

Appendix E – Affected Systems Studies

Appendix F – Deliverability Tables

Appendix G – Deliverability Results

Appendix H – Transmission Owner LPC Studies

Appendix I – Stability Plots

Appendix J – Short Circuit Studies